

P2247R1

2020 Library Evolution Report

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[GitHub](#)

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Table of Contents

- 1 Execution**
 - 1.1 Telecons
 - 1.2 Mailing List Reviews
 - 1.3 Electronic Polling
 - 1.4 Chair Guide
 - 1.5 Staff
- 2 Technical**
 - 2.1 Plenary Approved Priorities
 - 2.1.1 Executors
 - 2.1.2 Networking
 - 2.1.3 Coroutines Library Support

2.1.4 Standard Library Modules

2.2 Other Highlights

2.2.1 Formatting and Printing

2.2.2 Ranges

2.2.3 Text and Unicode

2.2.4 Freestanding

2.3 Other Work

References

Informative References

This report summarizes the activities of C++'s Library Evolution group from 2020-02 through 2020-10. It is split into two sections; one focused on *execution* (process, logistics, and operations) and one focused on *technical* work (papers we processed, etc).

Readers are encouraged to also look at:

- [\[P2145R1\]](#): Evolving C++ Remotely
- [\[P2138R3\]](#): Rules of Design $\Leftrightarrow$ Specification Engagement
- [\[P2195R0\]](#): Electronic Straw Polls

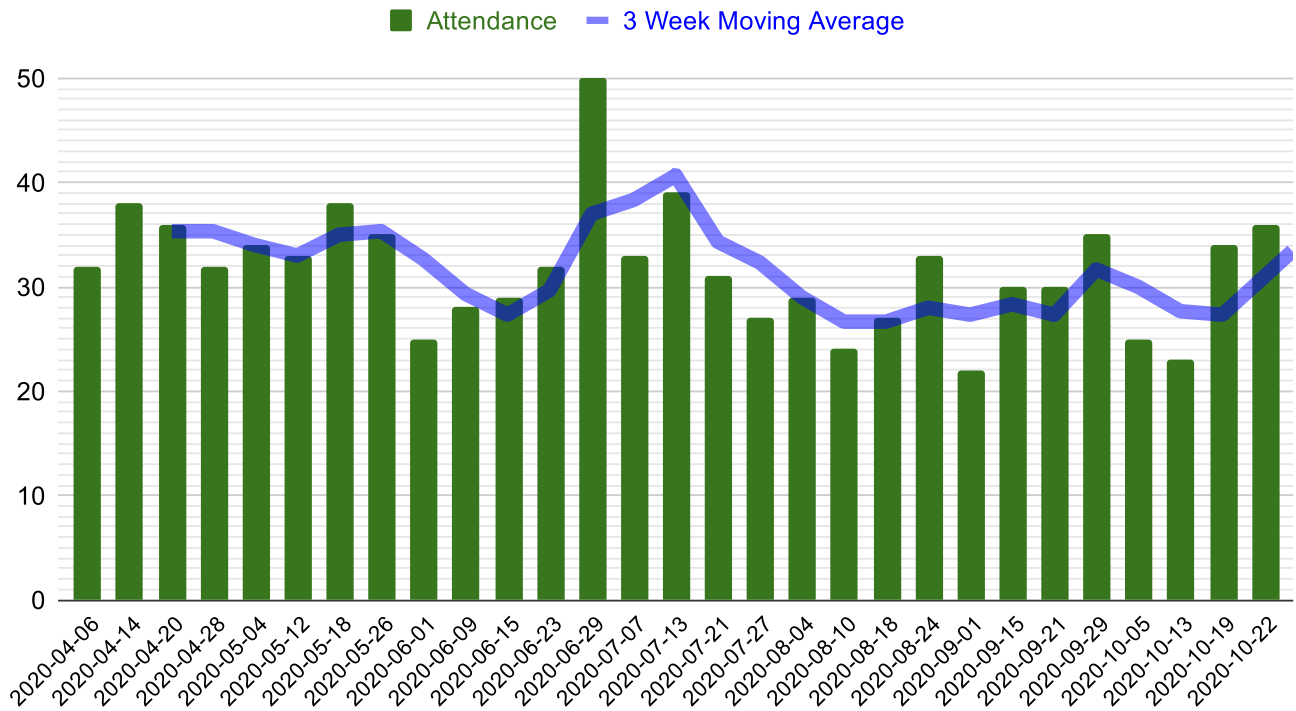
§ 1. Execution

§ 1.1. Telecons

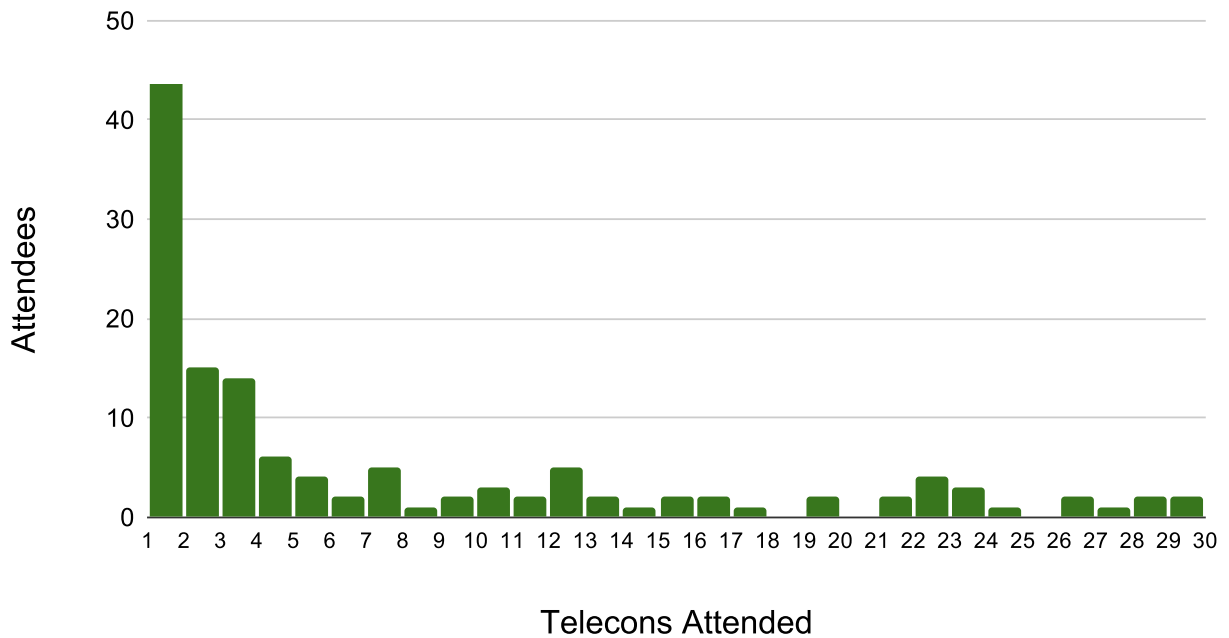
Since 2020-04, Library Evolution has held a 90 minute telecon twice a week to review papers. Typically, 1 or 2 papers are reviewed at each telecon.

More details on how we use telecons can be found in [\[P2145R1\]](#).

2020 Library Evolution Telecon Attendance



2020 Library Evolution Telecons Attended Per Attendee



# of Telecons	29
# of Papers Reviewed On Telecons	38
Mean Telecon Attendance	31.7
Total # of Attendees	132

Mean Telecons Per Attendee	7
Median Telecons Per Attendee	3

§ 1.2. Mailing List Reviews

Over the summer, we began conducting reviews of papers on the Library Evolution mailing list. We conduct 2 to 3 reviews concurrently, and each review lasts for a few weeks.

Mailing list reviews are critical to our operations, as they allow us to increase our paper-processing bandwidth without increasing the frequency or duration of telecons. Additionally, certain classes of papers tend to be particularly well suited for mailing list reviews, such as new proposals which tend need feedback, not decision making.

More details on how we use mailing lists can be found in [\[P2145R1\]](#).

# of Papers Reviewed On The Mailing List	14
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§ 1.3. Electronic Polling

In 2020-10, we began the polling period for our first slate of Library Evolution electronic straw polls. We believe electronic straw polls are the best option for Library Evolution to make decisions given our current process. They ensure that all stakeholders have an opportunity to participate.

More details on electronic polling can be found in [\[P2195R0\]](#).

§ 1.4. Chair Guide

We have developed a Library Evolution chairing guide, which can be found on the [Library Evolution GitHub wiki](#).

§ 1.5. Staff

Library Evolution is organized by a team, not a single individual:

- Bryce Adelstein Lelbach, Chair
- Fabio Fracassi, Vice Chair
- Ben Craig, Vice Chair
- Billy Baker, Incubator Chair
- Nevin Liber, Incubator Vice Chair
- Corentin Jabot, Mailing List Review Manager

— Inbal Levi, Mailing List Review Manager

Team leadership has a number of benefits:

- Increases our capacity and capabilities.
- Reduces the burden on each leader.
- Builds institutional knowledge and redundancy.

We hold a regular staff meeting every two weeks to discuss our operations and make plans for the future.

Additionally, the following noble individuals take minutes for Library Evolution, and deserve accolades:

- Ben Craig
- Inbal Levi
- Mark Hoemmen
- Guy Davidson

§ 2. Technical

§ 2.1. Plenary Approved Priorities

§ 2.1.1. Executors

During the Summer of 2020, Library Evolution conducted an extensive review of [\[P0443R13\]](#) Executors. Six review groups were formed, each of which focused on a particular aspect of the proposal and then wrote a report with their findings which was presented at a telecon:

- [\[P2183R0\]](#): Executors: Properties Review Report
- [\[P2202R0\]](#): Executors: Senders/Receivers Review Report
- [\[P2203R0\]](#): Executors: Customization Points Review Report
- [\[P2205R0\]](#): Executors: Polymorphic Executor Review Report
- [\[P2206R0\]](#): Executors: Static Thread Pool Review Report
- [\[P2207R0\]](#): Executors: Executor Concept Review Report

Additionally, we reviewed the following papers relating to Executors at telecons:

- [\[P2213R0\]](#): Executors Naming
- [\[P2220R0\]](#): Redefine Properties in P0443
- [\[P2221R0\]](#): Define P0443 CPOs with `tag_invoke`
- [\[P2235R0\]](#): Disentangling schedulers and executors

In 2020-10, we started taking electronic polls on Executors; the results are expected in 2020-11.

§ 2.1.2. Networking

Work on Networking has been proceeding in the Networking Study Group, not Library Evolution. We hope to begin a Library Evolution review of Networking for C++23 in the first half of 2021.

We did review [\[P2161R2\]](#) (Remove Default Candidate Executor), which proposes changes to the Networking Technical Specification. In 2020-10 we began electronic polling to advance this paper.

§ 2.1.3. Coroutines Library Support

In the Spring of 2020, we discussed [the roadmap for Coroutines Library Support](#) at a Library Evolution telecon. We also discussed the following papers at telecons:

- [\[P1288R0\]](#): Coroutine Concepts and Metafunctions
- [\[P1056R1\]](#): `std::lazy`
- [\[P2168R0\]](#): `std::generator`

No further work on Coroutines Library Support occurred after the Spring of 2020 due to a lack of new papers and a lack of revisions of existing papers. This plenary-approved priority is at risk of missing C++23 unless authors actively drive the work.

§ 2.1.4. Standard Library Modules

In the Spring of 2020, we discussed the existing proposal for Standard Library Modules ([\[P0581R1\]](#)) as well as a paper discussing the merits of modularizing the Standard Library ([\[P2172R0\]](#)).

No further work on Standard Library Modules occurred after the Spring of 2020 due to a lack of new papers and a lack of revisions of existing papers. This plenary-approved priority is at risk of missing C++23 unless authors actively drive the work.

§ 2.2. Other Highlights

§ 2.2.1. Formatting and Printing

We reviewed a proposed output counterpart to `std::format`, [\[P2093R1\]](#) (Formatted Output). The response was positive and we expect to see additional work in this area.

Additionally, a few papers proposing additions or fixes to `std::format` are on our agenda for 2020-11 telecons:

- [\[P2216R0\]](#): `std::format` improvements

— [\[P2197R0\]](#): Formatting for `std::complex`

§ 2.2.2. Ranges

In 2020-10, we will begin reviewing a group proposal suggesting a set of range features for C++23, [\[P2214R0\]](#), at Library Evolution telecons.

We reviewed the following ranges papers at telecons:

— [\[P1255R6\]](#): `std::views::maybe`

We reviewed the following ranges papers on the mailing list:

— [\[P2164R2\]](#): `std::views::enumerate`

§ 2.2.3. Text and Unicode

Work on Text and Unicode primarily occurs in the Text and Unicode Study Group, however a few papers are advancing to Library Evolution.

In 2020-11, we will begin a mailing list review of [\[P1885R3\]](#) Naming Text Encodings to Demystify Them.

§ 2.2.4. Freestanding

We reviewed a set of coupled papers proposing a reorganization and enhancement of freestanding on Library Evolution telecons:

— [\[P1641R3\]](#): Freestanding Library: Rewording the Status Quo

— [\[P1642R3\]](#): Freestanding [utilities], [ranges], and [iterators]

In 2020-11, we'll be reviewing a revision of [\[P1642R4\]](#) at a telecon.

§ 2.3. Other Work

The following papers were reviewed at Library Evolution telecons and in 2020-10 we started electronic polls to forward them to Library Working Group:

— [\[P2166R1\]](#): A Proposal to Prohibit `string` and `string_view` construction from `nullptr`.

— [\[P2212R1\]](#): Relax Requirements for `time_point::clock`

The following papers were reviewed at Library Evolution telecons:

— [\[LWG1396\]](#): `<regex>` should support allocators

- [\[LWG3430\]](#): `std::fstream` & co. should be constructible from `string_view`
- [\[LWG3440\]](#): Aggregate-paren-init breaks direct-initializing a tuple or optional from {aggregate-member-value}
- [\[LWG3451\]](#): Inconsistently explicit deduction guides
- [\[LWG3454\]](#): `pointer_traits::pointer_to` should be `constexpr`
- [\[P1467R4\]](#): Extended floating-point types
- [\[P1478R3\]](#): Byte-wise atomic memcpv
- [\[P1673R2\]](#): BLAS Linear algebra
- [\[P2037R1\]](#): String's gratuitous assignment
- [\[P2098R1\]](#): `std::is_specialization_of`
- [\[P2136R0\]](#): `std::invoke<R>`
- [\[P2139R1\]](#): Reviewing Deprecated Facilities for C++23
- [\[P2162R1\]](#): Inheriting from `std::variant` (resolving LWG3052)

The following papers were reviewed on the mailing list:

- [\[P1990R1\]](#): Add `operator[]` to `std::initializer_list`
- [\[P2165R1\]](#): Compatibility between tuple and tuple-like objects
- [\[P0870R4\]](#): A proposal for a type trait to detect narrowing conversions
- [\[P0205R0\]](#): Allow Seeding Random Number Engines With `std::random_device`
- [\[P1709R2\]](#): Graph Data Structures
- [\[P0952R0\]](#): A new specification for `std::generate_canonical`
- [\[P2146R2\]](#): Modern `std::byte` stream IO for C++
- [\[P1944R1\]](#): Add `constexpr` Modifiers to Functions in `<cstring>` and `<wchar>`
- [\[P1924R0\]](#): Making `std::stack` `constexpr`
- [\[P1925R0\]](#): Making `std::queue` `constexpr`
- [\[P1926R0\]](#): Making `std::priority_queue` `constexpr`

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